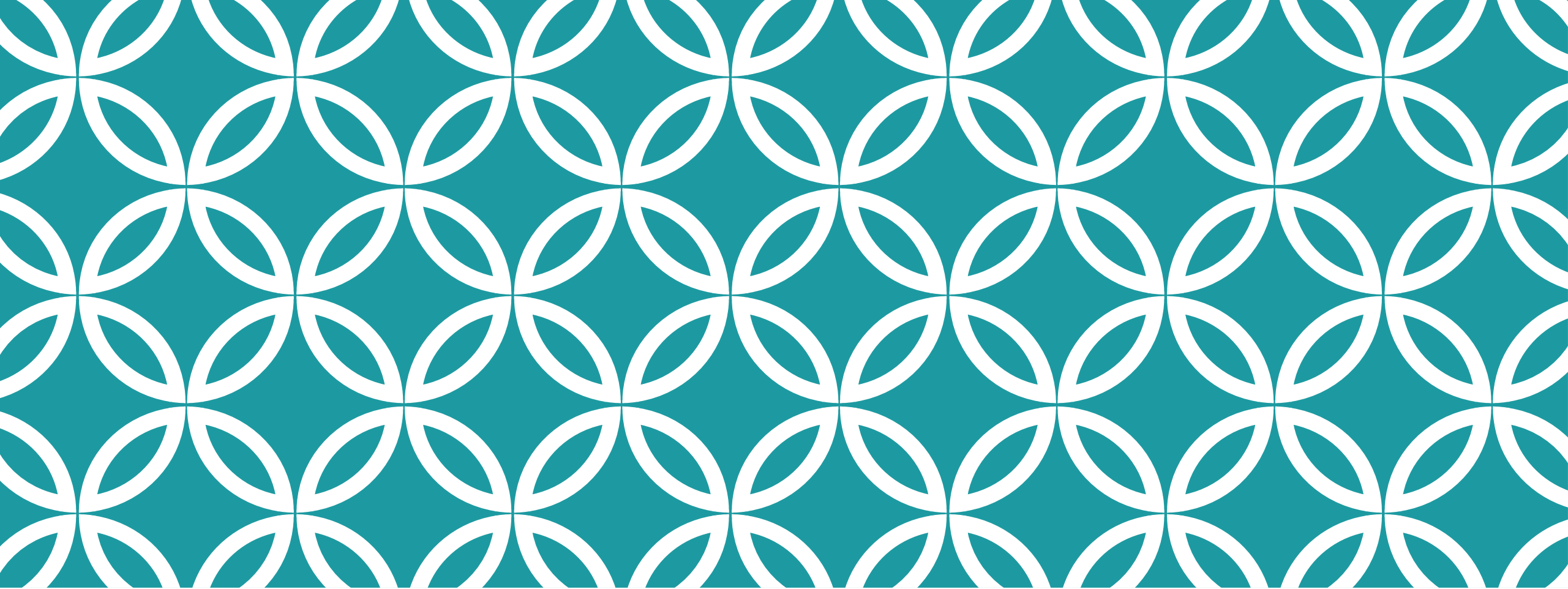


STAT 221- FUNDAMENTALS OF STATISTICAL DATA ANALYSIS

Week-2: Concepts Variables and
Measurement

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CONCEPTS, VARIABLES, AND MEASUREMENT



MEASUREMENT IN SCIENCE AND SOCIAL SCIENCE

Length, Area, Volume

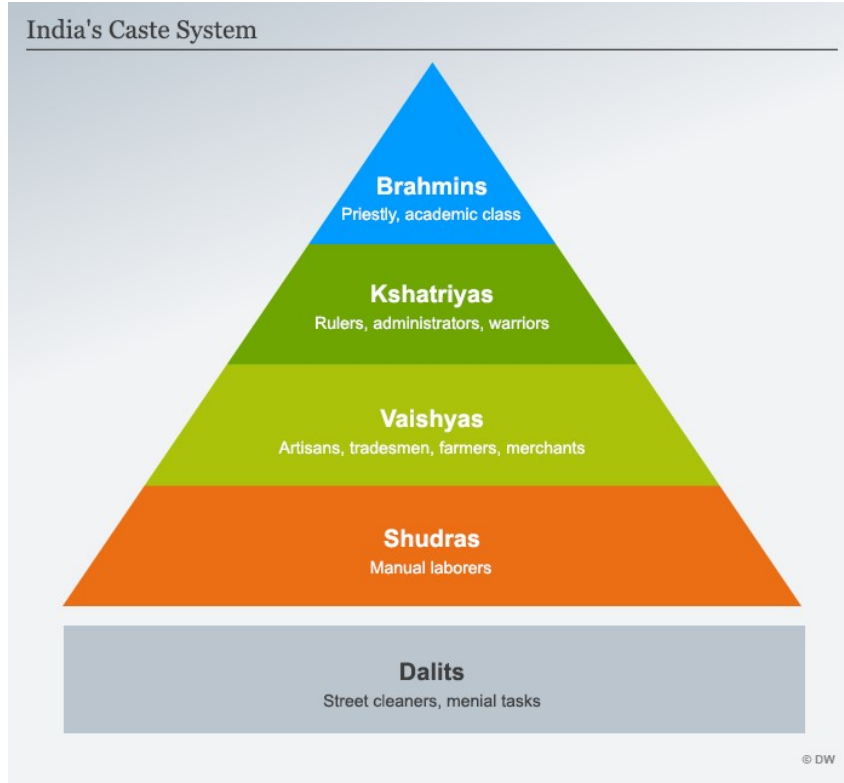
Science:

- ▢ Physics and Chemistry: Pressure, Volume, Temperature, time,
- ▢ Biology: Cell density, size, DNA seq; Blood counts- RBC, WBC

Social Science: **People, Society, Culture, Politics, Economy**

- ▢ Age, sex, weight, residence, place of birth, mother tongue
- ▢ Firm size, labour force, price of commodities
- ▢ Perceptions, opinions, creativity, **prejudice, discrimination**, happiness
- ▢ Wealth, National GDP, Assets
- ▢ Stratification, Power relation, Power distance, ...
- ▢ Libertarian, Authoritarianism, Inclusiveness ,... ,

CONCEPT, VARIABLE, AND MEASUREMENT



Caste System- and Castism Concept

How do we define it
?
How do we
measure it?

<https://www.dw.com/en/indias-caste-system-weakened-but-still-influential/a-39718124>

TERM	DEFINITION
Variable	Anything that varies from case to case <i>or</i> a logical set of attributes
Attribute	A characteristic of a variable

Conceptualization	The process of defining what is meant by a variable
Operationalization	The process of creating an actual measure for a variable

Nominal variable	A variable for which the attributes cannot be ranked from high to low
Ordinal variable	A variable for which the attributes can be ranked from high to low, but do not take numeric form
Interval/ratio variable	A variable for which the attributes can be ranked from high to low because they take numeric form

DEFINITIONS: CONCEPT, UNIT OF ANALYSIS, VARIABLE, ATTRIBUTES

Concept: An idea that we think represents something in the real world.
[Castism, Racism, Educational attainment, wealth, health]

Variable: Anything that may vary from one case to the next
□ [sex, height, level of education, etc.]

Attributes: A logical set of characteristics for a variable.
□ [Male/Female, (10 cm to 200 cm), primary/secondary/higher secondary/college]

Units of analysis: The who or what that data describe. Tend to consist of individuals or groups/places [Household, towns, schools, teams, etc.].

CONCEPTUALIZATION

Conceptualization is the mental process by which fuzzy and imprecise constructs (concepts) and their constituent components are defined in concrete and precise terms.

- ▮ **Prejudice:** Kind? What level?

(The process of understanding what is included and what is excluded in the concept of prejudice is the conceptualization process)

- ▮ **Discrimination**– Same as prejudice? Result of Prejudice?

- ▮ **Compassion** -- Empathy, sentimentality?

Dimensions

- ▮ **Unidimensional:** age, sex, weight,-- self esteem; relative deprivation of wealth;

- ▮ **Multidimensional:** Emotions (happiness, sadness, anger, fear, disgust); **Aptitude**- scientific , mathematical, verbal, artistic

OPERATIONALIZATION

Operationalization refers to **the process of developing indicators or items for measuring the conceptualized constructs.**

▢ Construct: Can have multiple indicators (Wealth, health, education...)

Indicators operate at the empirical level, in contrast to constructs, which are conceptualized at the theoretical level.

▢ **Reflective indicators:** A measure that “reflects” an underlying construct. (**attending religious services** may be a reflective indicator of religiosity)

▢ **Formative :** “forms” or contributes to an underlying construct (**belief dimension, a devotional dimension, and a ritual dimension**)

Variable: Each indicator- can have multiple **attributes** (levels) – carrying values; (Gender, Customer Satisfaction, Grade, Percentage)

Values of attributes may be **quantitative (numeric) or qualitative (non-numeric)**

LEVELS OF MEASUREMENT

On the Theory of Scales of Measurement

S. S. Stevens

TABLE 1

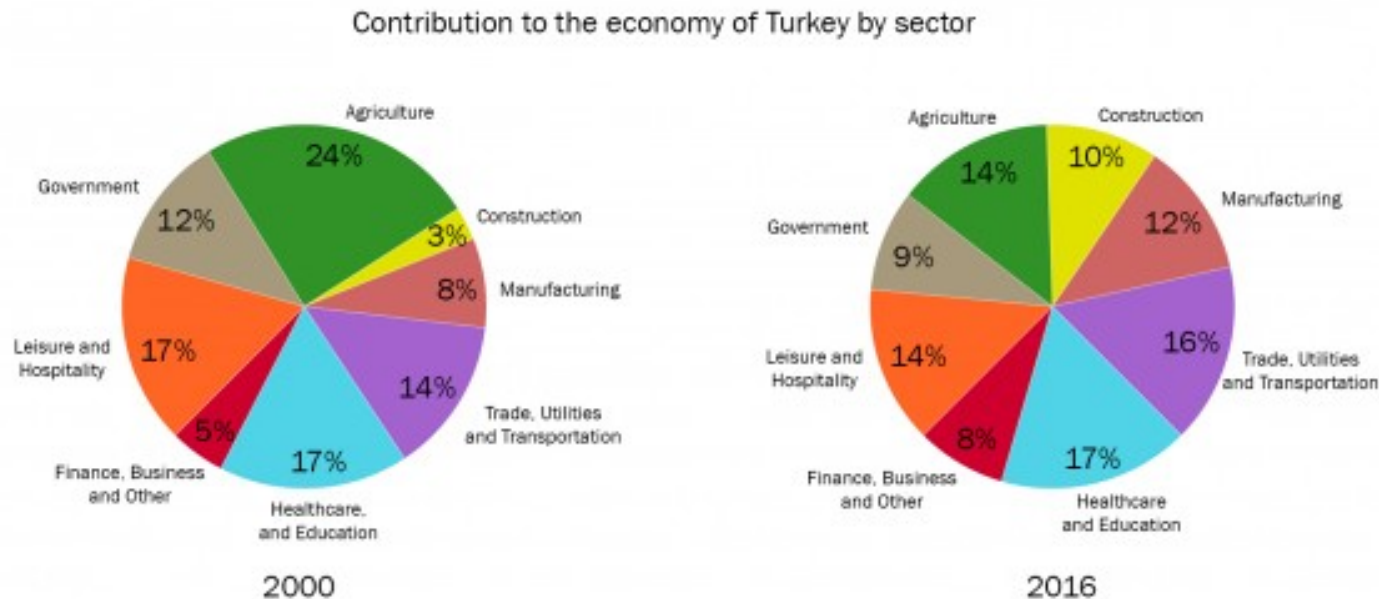
Scale	Basic Empirical Operations	Mathematical Group Structure	Permissible Statistics (invariantive)
NOMINAL	Determination of equality	<i>Permutation group</i> $x' = f(x)$ $f(x)$ means any one-to-one substitution	Number of cases Mode Contingency correlation
ORDINAL	Determination of greater or less	<i>Isotonic group</i> $x' = f(x)$ $f(x)$ means any monotonic increasing function	Median Percentiles
INTERVAL	Determination of equality of intervals or differences	<i>General linear group</i> $x' = ax + b$	Mean Standard deviation Rank-order correlation Product-moment correlation
RATIO	Determination of equality of ratios	<i>Similarity group</i> $x' = ax$	Coefficient of variation

NOMINAL SCALE

Nominal scales, **also called categorical scales, measure categorical data**. (Used for variables or indicators that have mutually exclusive attributes)

Gender (two values: male or female), industry type (manufacturing, financial, agriculture, etc.), and religious affiliation (Hindu, Muslim, Christian, Jew, etc.).

Anol Bhattacharjee,
2012



ORDINAL SCALE

Ordinal scales are those that measure rank-ordered data ($a_1 < a_2 < a_3, \dots < a_n$)

Ranking of students in a class as first, second, third, and so forth, based on their grade point average or test scores.

Can also use attribute labels(anchors) such as

- “bad”, “medium”, and “good”,
- **"strongly dissatisfied", "somewhat dissatisfied", "neutral", or "somewhat satisfied", and "strongly satisfied"**.

The central tendency measure of an ordinal scale can be its **median** or **mode**; percentiles correlation, regression, and analysis of variance, are not appropriate.

INTERVAL SCALES

Interval scales are those where **the values measured are not only rank-ordered but are also (usually) equidistant from adjacent attributes.**

Marks: { [0-10),[10,20),[20-30),...}, Temperature distribution; Income brackets

▢ What is the appropriate grouping? What scale you need to look at?

Allowed central tendency measures include **mean, median, or mode, as are measures of dispersion, such as range and standard deviation.**

Permissible statistical analyses include all of those allowed for nominal and ordinal scales, plus **correlation, regression, analysis of variance**, and so on.

RATIO SCALES

Ratio scales are those that have all the qualities of nominal, ordinal, and interval scales, and in addition, **also have a “true zero” point** (where the value zero implies lack or non-availability of the underlying construct);

- ▢ Natural science; Mass (kg), length(cm), pressure (Pa)
- ▢ Social science: Age, population size, Fertility rate, income, commodity price, firm size,...

All measures of central tendencies, including geometric and harmonic means, are allowed for ratio scales, as are ratio measures, such as studentized range or coefficient of variation. All statistical methods are allowed.

EXAMPLES AND EXERCISES

Interval variable

Ordinal variable Levels of education attended

- i. No schooling (0 years)
- ii. Primary (1 to 5th grade)
- iii. Upper Primary (6th to 8th grade)
- iv. Secondary (9th- 10th)
- v. Higher secondary (11th- 12th)/
Diploma/Vocational education
- vi. College
- vii. Post graduate

TABLE 12 *Example of an Ordinal Variable from the GSS*

DEGREE		RS HIGHEST DEGREE		
% Valid	% All	N	Value	Label
23.2	23.1	11,777	0	LT HIGH SCHOOL
51.7	51.6	26,307	1	HIGH SCHOOL
5.1	5.1	2,601	2	JUNIOR COLLEGE
13.6	13.6	6,918	3	BACHELOR
6.4	6.4	3,253	4	GRADUATE
	0.1	29	8	DK
	0.3	135	9	NA
100.0	100.0	51,020		Total

TABLE 13 *Example of an Interval/Ratio Variable from the GSS*

EDUCATION		HIGHEST YEAR OF SCHOOL COMPLETED		
% Valid	% All	N	Value	Label
0.3	0.3	137	0	
0.1	0.1	38	1	
0.3	0.3	132	2	
0.4	0.4	223	3	
0.6	0.5	280	4	
0.7	0.7	371	5	
1.3	1.3	666	6	
1.6	1.6	811	7	
4.9	4.8	2,470	8	
3.5	3.5	1,765	9	
4.8	4.8	2,451	10	
6.0	6.0	3,065	11	
31.1	31.0	15,814	12	
8.3	8.3	4,235	13	
10.7	10.6	5,422	14	
4.3	4.3	2,211	15	
11.8	11.8	6,025	16	
2.9	2.9	1,482	17	
3.3	3.3	1,660	18	
1.3	1.3	648	19	
1.9	1.9	962	20	
	0.1	64	98	DK
	0.2	88	99	NA
100.0	100.0	51,020		Total

Operationalizing Studiousness

We begin with the concept of *studiousness*. Studious can refer to many different ideas: attending class regularly, studying outside of class time, or doing all of the assigned readings. Suppose that we define it as the number of hours a person spends studying outside of class each week. We can measure this concept on a questionnaire in many different ways.

Consider the following:

Nominal: Do you study in the library? Yes No

Ordinal: How often do you study in the library? Never Sometimes Often

Interval/Ratio: How many hours do you spend studying in the library each week? _____

Operationalizing Course Challenge

Variables can be conceptualized and operationalized in any number of ways.

Nominal: Did you find the course challenging? Yes No

Ordinal: How challenging was the course? Not at all Somewhat Very

Interval/Ratio: In a range from 0 to 100, how challenging was this course? ____

1. The number of hours you spend studying each day (0, 1, 2, 3, 4, 5, . . .).
2. How often you would say you study (none, some, a lot).
3. Your favorite subject in school.
4. The frequency of skipping classes (never, sometimes, often).
5. The number of classes you skipped last semester (0, 1, 2, 3, 4,)
6. The number of study group meetings you had last semester (0–1, 2–3, 4–5, 6+)

TABLE 10 *Gun Control as a Nominal, Ordinal, and Interval/Ratio Variable*

NOMINAL	ORDINAL	RATIO
Does your state have strict gun control laws?	How important of an issue is gun control to you?	On a scale of 0 to 10, how important is gun control to you?
Yes No	Very important Somewhat important Not important	Respondent indicates a number.

EXERCISES

2. Suppose you want to operationalize racism at the individual level (the degree of racist opinions a person holds). Operationalize this concept in two ways.
3. Come up with a way to operationalize proficiency in Spanish.
7. What is the problem with the variable below?

THE NUMBER OF BOOKS YOU READ LAST SUMMER

0–2 books

2–4 books

4–6 books

6 or more books

**Mutually exclusive
collectively
exhaustive**

8. What is the problem with the variable below?

THE NUMBER OF BOOKS YOU READ LAST SUMMER

1–2 books

3–5 books

6 or more books

At what level are each of the following variables operationalized?

9. *Age* (1, 2, 3 . . .)
10. *Sex* (male, female, other)
11. *Class standing* (freshman, sophomore, junior, senior, other)
12. *Marital status* (married, single, divorced, widowed, other)
13. *Median Household Income* (annual household income in dollars)
14. How much you like the food on campus (a lot, some, a little, other)
15. The number of toxic waste sites in your community (0, 1, 2, 3, . . .)
16. The number of toxic waste sites in your community (0, 1–5, 6–10, 11+)
17. *Your GPA* (below average, average, above average)
18. Explain why it is best to collect data at the highest level of measurement possible.

INDIVIDUAL AND ECOLOGY FALLACY IN DATA

Individual Fallacy

Individual fallacy is a generalization from an individual (or small group) to a large group

Ex:

- Individual taste/opinion of food or politics to a whole group
- Attributing crime committed by a person, to the characteristics of whole groups
- Generalization from anecdotal evidences

Ecological fallacy

Ecological fallacy is a mistake that is made when we impose characteristics of a group upon individuals in that group.

Ex:

- Imposing group statistics on crime on individual from groups
- Gender stereotyping on individuals
- Castist remarks on education and intelligence of individual

SCALE AND STATISTICS

Scale	Central Tendency	Statistics	Transformations
Nominal	Mode	Chi-square	One-to-one (equality)
Ordinal	Median	Percentile, non-parametric statistics	Monotonic increasing (order)
Interval	Arithmetic mean, range, standard deviation	Correlation, regression, analysis of variance	Positive linear (affine)
Ratio	Geometric mean, harmonic mean	Coefficient of variation	Positive similarities (multiplicative, logarithmic)
Note: All higher-order scales can use any of the statistics for lower order scales.			

RATING SCALES - ORDINAL

Common rating scales include **binary**, **Likert**, **semantic differential**, or **Guttman scales**.

Binary scale: Yes/No; True/False

Political Activism	BINARY SCALE	
	Yes	No
Have you ever written a letter to a public official	Yes	No
Have you ever signed a political petition	Yes	No
Have you ever donated money to a political cause	Yes	No
Have you ever donated money to a candidate running for public office	Yes	No
Have you ever written a political letter to the editor of a newspaper or magazine	Yes	No
Have you ever persuaded someone to change his/her voting plans	Yes	No

LIKERT SCALE

	Strongly Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Strongly Agree
I feel good about my job	1	2	3	4	5
I get along well with others at work	1	2	3	4	5
I'm proud of my relationship with my supervisor at work	1	2	3	4	5
I can tell that other people at work are glad to have me there	1	2	3	4	5
I can tell that my coworkers respect me	1	2	3	4	5
I feel like I make a useful contribution at work	1	2	3	4	5

Table 6.3. A six-item Likert scale for measuring employment self-esteem

SEMANTIC DIFFERENTIAL SCALE.

This is a composite (multi-item) scale **where respondents are asked to indicate their opinions or feelings toward a single statement** using **different pairs of adjectives framed as polar opposites**.

How would you rate your opinions on national health insurance?						
	Very much	Somewhat	Neither	Somewhat	Very much	
Good	☐	☐	☐	☐	☐	Bad
Useful	☐	☐	☐	☐	☐	Useless
Caring	☐	☐	☐	☐	☐	Uncaring
Interesting	☐	☐	☐	☐	☐	Boring

Table 6.4. A semantic differential scale for measuring attitude toward national health insurance

Bhattacharjee, A. (2012). Social science research: Principles, methods,

GUTTMAN SCALE

This composite scale **uses a series of items arranged in increasing order of intensity of the construct of interest**, from least intense to most intense.

Each item in Guttman scale **has a weight**

How will you rate your opinions on the following statements about immigrants?		
Do you mind immigrants being citizens of your country	Yes	No
Do you mind immigrants living in your own neighborhood	Yes	No
Would you mind living next door to an immigrant	Yes	No
Would you mind having an immigrant as your close friend	Yes	No
Would you mind if someone in your family married an immigrant	Yes	No

**Replace
immigrant
by - other
religion**

Table 6.5. A five-item Guttman scale for measuring attitude toward immigrants

Bhattacharjee, A. (2012). Social science research: Principles, methods, and practices.